15. Exercise distribution

10

Easy

10

Medium

5

Hard / Tricky

1

Scenario

26 exercises total, progressing from small functions to an integrated object-oriented application.

Level 1 – Easy exercises

No.	Exercise	Main concept
E01	Employee Greeting Function	Function definition and invocation
E02	Annual Salary Calculator	Parameters and return values
E03	Grade Classification Function	Conditional logic in functions
E04	Employee Profile Generator	Multiple arguments
E05	Configurable Bonus Calculator	Default arguments
E06	Assessment Average Calculator	`*args`
E07	Employee Metadata Printer	`**kwargs`
E08	Employee ID Validator Class	Classes and objects
E09	Training Course Class	Constructors and instance methods
E10	Simple Student Inheritance	Base and derived classes

Level 2 – Medium exercises

No.	Exercise	Main concept
M01	Employee Record Management Functions	Typed functions and dictionaries
M02	Department Salary Analyzer	Multiple function arguments and collections
M03	Employee with Validated Assessment Score	Encapsulation and property

M04	Permanent and Contract Employee Models	Inheritance and method overriding
M05	Training Notification System	Duck typing and polymorphism
M06	Certification Policy Engine	Abstract classes
M07	Training Record Composition	Has-a relationships
M08	Function Execution Logger	Decorators and `wraps`
M09	Performance Timer	Decorators with arguments
M10	Employee Report Service	Class methods, static methods and separation of concerns

Level 3 – Hard / Tricky exercises

No.	Exercise	Challenge
H01	Custom Role-Based Decorator	Decorator factory and function wrapping
H02	Multiple Inheritance Employee Model	MRO and `super()`
H03	Type-Safe Generic Data Processor	Generic types and annotations
H04	Stateful Function Counter	Closures and `nonlocal`
H05	Extensible Certification Engine	Abstract classes, polymorphism, composition and decorators

Level 4 – Integrated scenario

H06 – Enterprise application

Employee Training and Certification Management Platform

An organization manages employees, training programmes, certification policies and employee performance.

Develop an application supporting:

- Employee creation and validation.
- Multiple employee categories.
- Reusable business functions.
- Employee registration and lookup.
- Standard and advanced certification rules.
- Role-based report access.
- Execution logging through decorators.
- Performance and department-wise reporting.
- Exception handling.
- Type annotations and automated tests.

Technical constraints:

- Use classes, inheritance, encapsulation and polymorphism.
- Use functions with suitable parameter and return types.
- Use at least two decorators.
- Use an abstract base class or another well-defined common interface.
- Use composition where appropriate.
- Do not place all application logic in one class.
- Run Ruff, mypy and Bandit.
- Write at least five meaningful automated tests.

Expected deliverable: A small modular Python application rather than a single procedural script.